FIRST and FOLLOW

$S \rightarrow ABCD$

$A \rightarrow a / \varepsilon$

$B \rightarrow b / \varepsilon$

$C \rightarrow AB / c$

$D \rightarrow d$

FIRST

$FIRST(S) = \{\$$

$FIRST(D) = \{d\}$

$FIRST(B) = \{b, \varepsilon\}$

$FIRST(A) = \{a, \varepsilon\}$

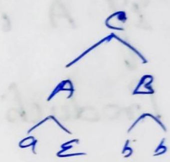
$FIRST(C) = FIRST(AB) + FIRST(c)$

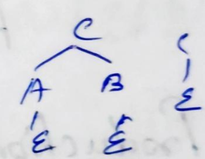
$= \{a, \varepsilon\}$

$\varepsilon \rightarrow FIRST(B)$

$= \{a, b, \varepsilon\} + \{c\}$

$= \{a, b, c, \varepsilon\}$





$FIRST(\cancel{ABCD}\ S) = FIRST(ABCD)$

$= \cancel{FIRST} \{a, \varepsilon\}$

$\varepsilon \rightarrow FIRST(BCD)$

$= \{a, b, \varepsilon\}$

$\varepsilon \rightarrow FIRST(CD)$

$= \{a, b, c, \varepsilon\}$

$\varepsilon \rightarrow FIRST(D)$

$= \{a, b, c, d\}$

FIRST

② $E \to TE'$
$E' \to +TE' / \epsilon$
$T \to FT'$
$T' \to *FT' / \epsilon$
$F \to (E) / id$

$\text{FIRST}(E) = \text{FIRST}(T) = \text{FIRST}(F)$
$= \{(, id\}$
$\text{FIRST}(E') = \{+, \epsilon\}$
$\text{FIRST}(T') = \{*, \epsilon\}$

→ Compute follow

(i) If A is a start symbol, put \$ in FOLLOW(A)

(ii) Production of the form $B \to \alpha A \beta$

$$\text{FOLLOW}(A) = \text{FIRST}(\beta)$$

(iii) Production of the form $B \to \alpha A$ or $B \to \alpha A \beta$, where $\beta \xRightarrow{*} \epsilon$, add

$$\text{FOLLOW}(A) = \text{FOLLOW}(B)$$

$\text{FOLLOW}(E) = \{\$,)\}$

$\text{FOLLOW}(E') = \{\text{FOLLOW}(E') + \text{FOLLOW}(E)$
$= \{\$,)\}$

$\text{FOLLOW}(T) = \text{FIRST}(E')$
$= \{+, \epsilon\}$
$\quad \hookrightarrow \text{FOLLOW}(E)$
$= \{+, \$,)\}$

$\text{FOLLOW}(T') = \{\text{FOLLOW}(T') + \text{FOLLOW}(T)$
$= \{+, \$,)\}$

$\text{FOLLOW}(F) = \{*, +, \$,)\}$

$\alpha \; A \; \beta$
$F \to (E)$
$\text{FOLLOW}(E) = \text{FIRST}(\beta)$

$E \to TE'$
$E' \to +TE'$

$E \to TE'$
$E' \to +TE'$

$\text{FOLLOW}(E') = \text{FOLLOW}(E)$
(Bcoz no symbol after E')

$\text{FOLLOW}(E') = \text{FOLLOW}(E')$

③ $S \rightarrow (A) \mid \epsilon$

$A \rightarrow TE$

$E \rightarrow \& TE \mid \epsilon$

$T \rightarrow (A) \mid a \mid b \mid c$

$\text{FIRST}(S) = \{(, \epsilon\}$ $\quad S \rightarrow \underline{(}A) \mid \underline{\epsilon}$

$\text{FIRST}(A) = \{(, a, b, c\}$ $\quad \left\{ \begin{array}{l} A \rightarrow TE \\ T \rightarrow \underline{(}A) \mid \underline{a} \mid \underline{b} \mid \underline{c} \end{array} \right\}$

$\text{FIRST}(E) = \{\&, \epsilon\}$ $\quad E \rightarrow \underline{\&}TE \mid \underline{\epsilon}$

$\text{FIRST}(T) = \{(, a, b, c\}$ $\quad T \rightarrow \underline{(}A) \mid \underline{a} \mid \underline{b} \mid \underline{c}$

$\text{FOLLOW}(S) = \{\$\}$

$\text{FOLLOW}(A) = \{)\}$ $\quad A \rightarrow TE$

$\text{FOLLOW}(E) = \text{FOLLOW}(A) = \{)\}$ $\quad E \rightarrow \& TE \mid \epsilon$

$\text{FOLLOW}(T) = \text{FIRST}(E)$ $\quad A \rightarrow T\underline{E}$

$= \{\&, \epsilon\}$ $\quad E \rightarrow \& T\underline{E}$

$\epsilon \hookrightarrow \text{FOLLOW}(E)$

$= \{\&,)\}$

$A \rightarrow a / \varepsilon$

$B \rightarrow b / \varepsilon$

$C \rightarrow c$

$D \rightarrow d / \varepsilon$

$E \rightarrow e / \varepsilon$

FIRST (A) = {a, ε}

FIRST (B) = {b, ε}

FIRST (C) = {c}

FIRST (D) = {d, ε}

FIRST (E) = {e, ε}

FIRST (S) = FIRST(ABCD)

= {a, ε} ↳ FIRST (BCD)

= {a, b, ε} ↳ FIRST(CD)

= {a, b, c, ~~ε~~} ~~↳ FIRST(D)~~

= ~~{a, b, c, d, ε}~~

= ~~{a, b, c, d, e}~~

S → A B C D
(A may be a, ε)
if ε then
FIRST (BCD)

FOLLOW (S) = {\$}

FOLLOW (A) = FIRST (B)

$A \overset{\varepsilon}{\cancel{B}} C D E$

= {b, ε} ↳ FIRST(C)

= {b, c}

FOLLOW (B) = FIRST (C) = {c}

FOLLOW (C) = FIRST (D) = {d, ε} = {d, e, ε} = {d, e, \$}

↳ FIRST (E) ↳ FOLLOW(S)

FOLLOW (D) = FIRST (E) = {e, ε} = {e, \$}

FOLLOW (E) = FOLLOW (S) = {\$}

5. $A \rightarrow aBa$

$B \rightarrow bB / \varepsilon$

FOLLOW$(A) = \{\$\}$

FOLLOW$(B) = \{a\}$

FIRST$(A) = \{a\}$

FIRST$(B) = \{b, \varepsilon\}$

6. $S \rightarrow Bb / Cd$

$B \rightarrow aB / \varepsilon$

$C \rightarrow cC / \varepsilon$

FIRST$(B) = \{a, \varepsilon\}$

FIRST$(C) = \{c, \varepsilon\}$

FIRST$(S) =$ FIRST(Bb) + FIRST(Cd)

$= \{a, b, c, d\}$

FOLLOW$(S) = \{\$\}$

FOLLOW$(B) = \{b\}$

FOLLOW$(C) = \{d\}$

7. $S \rightarrow ACB \mid CbB \mid Ba$

$A \rightarrow da \mid BC$

$B \rightarrow g \mid \varepsilon$

$C \rightarrow h \mid \varepsilon$

FIRST $(B) = \{g, \varepsilon\}$

FIRST $(C) = \{h, \varepsilon\}$

FIRST $(A) = \{d, g, h, \varepsilon\}$ (FIRST (B), FIRST (C))

FIRST (S) = FIRST (ACB) + FIRST (CbB) + FIRST (Ba)

$= \{d, g, h, \varepsilon\} + \{h, b\} + \{g, a\}$

$= \{d, g, h, a, b, \varepsilon\}$

FOLLOW $(S) = \{\$\}$

FOLLOW $(A) = \{$ FIRST (C)

$= \{h, \varepsilon\}$ ↳ FIRST (B)

$= \{h, g, \varepsilon\}$ ↳ FOLLOW (S)

$= \{h, g, \$\}$

FOLLOW $(B) = \{\$, a, h, \varepsilon\}$ ↳ FOLLOW (A)

$= \{\$, a, h, g\}$

FOLLOW $(C) = \{b, g, \$, b, a, h\}$

$= \{g, \$, b, h\}$

FIRST function

1. for a produce rule $X \rightarrow \varepsilon$,

$$FIRST(X) = \{\varepsilon\}$$

2. for any terminal symbol 'a'

$$FIRST(a) = \{a\}$$

3. for a production rule $X \longrightarrow Y_1 Y_2 Y_3$

* Calculating $FIRST(X)$
  * If $\varepsilon \notin FIRST(Y_1)$, then $FIRST(X) = FIRST(Y_1)$
  * If $\varepsilon \in FIRST(Y_1)$, then

$$FIRST(X) = \{FIRST(Y_1) - \varepsilon\} \cup FIRST(Y_2 Y_3)$$

* Calculating $FIRST(Y_2, Y_3)$
  → If $\varepsilon \notin FIRST(Y_2)$, then $FIRST(Y_2 Y_3) = FIRST(Y_2)$
  → If $\varepsilon \in FIRST(Y_2)$, then

$$FIRST(Y_2 Y_3) = \{FIRST(Y_2) - \varepsilon\} \cup FIRST(Y_3)$$

FOLLOW function

1. for the start symbol S, place \$ in FOLLOW(S)
2. for any production rule $A \rightarrow \alpha B$

$$\text{FOLLOW}(B) = \text{FOLLOW}(A)$$

3. for any production rule $A \rightarrow \alpha B \beta$

* If $\epsilon \notin \text{FIRST}(\beta)$, then $\text{FOLLOW}(B) = \text{FIRST}(\beta)$
* If $\epsilon \in \text{FIRST}(\beta)$, then

$$\text{FOLLOW}(B) = \{\text{FIRST}(\beta) - \epsilon\} \cup \text{FOLLOW}(A)$$

8. Calculate first and follow functions for the given grammar.

$S \rightarrow aBDh$
$B \rightarrow cC$
$C \rightarrow bC / \epsilon$
$D \rightarrow EF$
$E \rightarrow g / \epsilon$
$F \rightarrow f / \epsilon$

FIRST$(S) = \{a\}$
FIRST$(B) = \{c\}$
FIRST$(C) = \{b, \epsilon\}$
FIRST$(D) = \{(\text{FIRST}(E) - \epsilon) \cup \text{FIRST}(F)\}$
$= \{g, f, \epsilon\}$
FIRST$(E) = \{g, \epsilon\}$
FIRST$(F) = \{f, \epsilon\}$

FOLLOW$(S) = \{\$\}$
FOLLOW$(B) = \{\text{FIRST}(D) - \epsilon\} \cup \text{FIRST}(h)$
$= \{g, f, h\}$
FOLLOW$(C) = \text{FOLLOW}(B)$
$= \{g, f, h\}$
FOLLOW$(D) = \{h\}$
FOLLOW$(E) = \{\text{FIRST}(F) - \epsilon\} \cup \text{FOLLOW}(D) = \{f, h\}$
FOLLOW$(F) = \text{FOLLOW}(D)$
$= \{h\}$

9. Calculate the first and follow functions for the given grammar

$S \to A$

$A \to aB \mid Ad$

$B \to b$

$C \to g$

→ The given grammar is left recursive, so we remove left recursion.

$S \to A$

$A \to aBA'$

$A' \to dA' \mid \epsilon$

$B \to b$

$C \to g$

FIRST(S) = {a}

FIRST(A) = {a}

FIRST(A') = {d, ε}

FIRST(B) = {b}

FIRST(C) = {g}

follow(S) = {\$}

follow(A) = follow(S) = {\$}

follow(A') = follow(A) = {\$}

follow(B) = {d, \$}

follow(C) = NA

B A'

first(A')

(d, ε)

follow(A)

10. Calculate the first and follow functions for the given grammar

$S \to (L) / a$

$L \to SL'$

$L' \to ,SL' / \varepsilon$

$first(S) = \cancel{\{(\}}\ \{(, a\}$

$first(L) = first(S) = \{(, a\}$

$first(L') = \{,, \varepsilon\}$

SL'
$\downarrow$
$first(L')$
$,, \varepsilon$
$\downarrow$
$follow(L)$

$follow(S) = \cancel{\{\$, ,,)\}}\ \{\$, ,,)\}$

$follow(L) = \{)\}$

$follow(L') = \{)\}$

———— α ————

Remove left recursion

$S \to ABC$

$A \to Aa / Ad / b$

$B \to Bb / c$

$C \to Ce / g$

$A \to A\alpha / \beta$
$\Downarrow$
$A \to \beta A'$
$A' \to \alpha A' / \varepsilon$

$S \to ABC$

$A \to bA'$

$A' \to aA' / \varepsilon / dA'$

$B \to cB'$

$B' \to bB' / \varepsilon$

$C \to gC'$

$C' \to eC' / \varepsilon$